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December 14, 1972
SUPERSEDING
Int. Fed. Spec. RR-W-00420A(GSA-FSS)
August 24, 1970 and
Fed. Spec. RR-W-420
August 10, 1956

FEDERAL SPECIFICATION

WIRE ROPE, STEEL (HIGHWAY GUARD)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers zinc-coated steel wire ropes for highway guard rails.

1.2 Classification.

1.2.1 Types and classes. Wire rope furnished under this specification shall be of the following types and classes, as specified (see 6.2):

Type I - 3/4 inch (3 by 7).

Class 1 - 15,000-pound breaking strength.
Class 2 - 25,000-pound breaking strength.

Type II - 1 inch (7 by 7).

Class 2 - 25,000-pound breaking strength.
Class 3 - 45,000-pound breaking strength.

1.2.2 Length. Unless otherwise specified (see 6.2), the wire rope shall be furnished in continuous lengths of 500, 1000, or 2,000 feet as specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 4010

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Military Standard:

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Association of State Highway Official (AASHO) Publication:

AASHO T39 - Wire Rope (Guardrail), Method of Test for.

(Application for copies should be addressed to American Association of State Highway Officials, 341 National Press Building, Washington, DC 20004.)

3. REQUIREMENTS

3.1 Material. Wires in highway guard wire ropes shall be annealed steel made in open-hearth, basic-oxygen or electric furnaces.

3.2 Wire rope.

3.2.1 Type I. Type I wire rope shall consist of three strands laid around each other (see fig. 1).

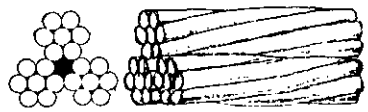


FIGURE 1. 3/4 inch (3 by 7) wire rope.

3.2.2 Type II. Type II wire rope shall consist of six outer strands laid spirally (helically) around a center core wire (see fig. 2).

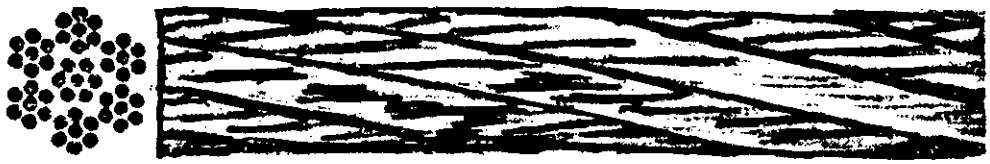


FIGURE 2. 1 inch (7 by 7) wire rope.

3.2.3 Form. Unless otherwise specified (see 6.2), wire rope shall be preformed or non-preformed at the manufacturer's option.

3.2.4 Strands. Each strand shall consist of outer wires laid spirally around a center wire (see fig. 1 and 2).

3.2.5 Lay. Wire rope shall be right regular lay. The wires in the strands shall be laid to left and each strand laid to the right (see fig. 1 and 2).

3.2.6 Pitch. Wire in strands and strands shall be closely laid with uniform pitches.

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3.2.7 Breaking strength. Breaking strength of wire rope shall be as follows (see 4.4.1):

Type	Class	Breaking strength minimum, pounds
1	1	15,000
1	2	25,000
2	2	25,000
2	3	45,000

3.3 Wire.

3.3.1 Zinc coat. Each wire shall be zinc coated. Weight of zinc shall be 0.8 ounces per square foot or greater (see 4.4.2).

3.3.2 Wrapping. Individual wires shall withstand the wrapping test in 4.4.3 without breakage or fracture of the steel.

3.4 Wire rope and wire.

3.4.1 Dimensions and number of wires in rope. Number of wires in rope, and dimensions of wire rope and individual wires shall be in accordance with table I.

TABLE I. Wire rope and wire in rope

Wire rope and wires	Type I 3/4 inch	Type II 1 inch
Number of wires in a wire rope	21	49
Wire diameter, minimum-maximum, inch	0.117-0.124	0.105-0.112
Pitch of wires in a strand, maximum, inch	4-1/2	4-1/2
Ordered (nominal) diameter, minimum, inch	3/4 inch	1 inch
Maximum diameter, inches	25/32	1-3/64
Pitch of strands in a wire rope, maximum, inches	7-1/2	10

3.4.2 Joints. When joints are used, the distance between joints in any two individual wires shall be not less than 18 inches.

3.4.2.1 Individual wires. When joints are used, the distance between them shall be 18 inches or greater.

3.4.2.2 Joining methods. Joining of individual wires shall be either by lap brazing or butt welding.

3.4.2.2.1 Lap joints. The lap in lap-brazed joints shall be not less than 3 diameters of the wires with the overlapping faces covered by brazing metal.

3.4.2.2.2 Butt welded joints. Wire ends shall not show any indications of overheating which may affect the properties of the wire. Flash shall be removed from joints.

3.4.2.2.3 Lap and butt joints. All joints shall be zinc-coated after welding.

3.5 Workmanship.

3.5.1 Wire rope. Wire rope shall be free from kinks, splits and loose wires.

3.5.2 Zinc coat on wires. Coating shall be free from uncoated spots, scales, lumps, pits or blisters.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling.

4.2.1 Lot. A lot shall consist of each type and class submitted for acceptance at one time.

4.2.1.1 When wire rope or strand can be identified by its manufacturing reel, one sample 12 feet long shall be taken from one shipping reel in the lot cut from the same manufacturing reel.

4.2.1.2 When wire rope or strand cannot be identified by its manufacturing reel, one sample 12 feet long shall be cut from each reel or coil submitted for inspection.

4.2.1.3 The 12-foot samples shall be seized, and cut into two pieces: A 3-foot length and a 9-foot length. Hereinafter the 3-foot length shall be identified as the short sample and the 9-foot length as the long sample.

4.2.1.4 Sampling for zinc coating. A minimum of four specimens of a convenient length, but not less than 12 inches each, cut from each size wire used in the rope, shall be randomly selected from the short samples.

4.3 Visual and dimensional examination. Each reel (or coil) in the lot shall be visually examined for workmanship, measured for dimensional requirements.

4.3.1 Examination of diameter. Using the free end and beginning not less than 5 feet from the end of the wire rope, measure the diameter of the wire rope using a caliper across the largest diameter.

4.4 Test methods.

4.4.1 Breaking strength. Determine breaking strength using the long specimens. Cut to suitable length and prepare ends of specimens and end anchors for testing in accordance with AASHO T39, except when cleaning wire do not use gasoline. Test specimens in any suitable tensile testing machine (see 3.2.7).

4.4.2 Zinc coat on wires.

4.4.2.1 Preparation of specimen. Wash with solvent naptha or other suitable solvent, then with alcohol, and dry thoroughly.

4.4.2.2 Weight of coating. Test for weight by the Hydrochloric Acid-Antimony Chloride Method also called the stripping method. Hydrochloric acid shall be CP or reagent grade.

4.4.2.3 Preparation of the stripping solution. Dissolve twenty grams (gm.) of antimony trioxide (Sb_2O_3) or 32 gm. of antimony trichloride ($SbCl_3$) in 1,000 milliliters (ml.) of concentrated hydrochloric acid (HCl) having a specific gravity of approximately 1.19. Add 5 ml. of the received antimony chloride solution to each 100 ml. of concentrated hydrochloric acid (HCl) having a specific gravity of approximately 1.19 used as a zinc stripping solution.

4.4.2.4 Procedure. Weigh the cleaned specimen to the nearest 0.01 gm.

(a) Strip the zinc coating from the weighted specimen by immersing it in the stripping solution until the evolution of hydrogen ceases, or until only few bubbles are evolved. If the vessel used for the stripping bath is of small size, loosely coil the specimen to facilitate complete immersion. Regulate temperature of the stripping solution so that it does not exceed 100°F. (38°C). The same solution may be repeatedly used without further additions of antimony choride solution, until the time for stripping becomes inconveniently long.

(b) After stripping wash and scrub the specimen under running water, and dry.

(c) Determine weight of the stripped wire to the nearest 0.01 gm.

(d) Determine diameter of stripped specimens, in inches, by taking the mean of two measurements at right angles to each other. Measure to the nearest 0.001 inch.

(e) Calculate the weight of zinc coating as follows (see 3.3.1):

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$$C = \frac{W_1 - W_2}{W_2} \times d \times M$$

Where:

C = Weight of zinc coating in onces per square foot of stripped wire surface
W_1 = Original weight of specimen in grams.
W_2 = Weight of stripped specimen in grams.
D = Diameter of stripped wire in inches or mm.
M = Constant equal to:

163 if "d" is in inches.
6.42 if "d" is in mm.

4.4.3 Wrapping test. Wrap either size specimen around a mandrel three times the diameter of the wire, and for six turns closely wound around the mandrel.

4.5 Rejection and retests.

4.5.1 Rejection. If any of the test specimens fail to pass any specified tests, all reels or coils from the same manufacturing reel or coil in the lot shall be rejected. If the test specimens were taken from each individual reel or coil the particular reel or coil, the specimen of which failed in the tests, shall be rejected.

4.5.2 Retest. The rejected material at the request from the manufacturer may be retested in accordance with Fed. Test Method Std. No. 151. Only one retest shall be permitted for each lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packing of highway guard wire rope shall be level A or C, as specified (see 6.2), and in lengths as specified (see 6.2), in accordance with MIL-STD-163.

5.2 Marking. In addition to any special marking required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129 or Fed. Std. No. 123, as applicable.

6. NOTES

6.1 *Intended use.* *Wire rope guard rail is intended for use along roads to restrain out-of-control vehicles and for temporary replacement of damaged formed steel guardrail.*

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class (see 1.2.1).
(c) Length (see 1.2.1).
(d) Form (see 3.2.3).
(e) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 Definitions.

6.3.1 Breaking strength. The measured load required to pull apart the wire rope in tension.

6.3.2 Wire Rope diameter (ordered or nominal). The measured diameter of the circle circumscribing all strands in the wire rope.

6.3.3 Lay. The manner in which the outer wires in a strand and the outer strands in a wire rope are laid (twisted) around a center (core) wire or strand respectively.

6.3.4 Pitch. The distance, parallel to the axis of the strain or wire rope, in which a wire or a strand make a complete turn. (The pitch is also called "the lay length").

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6.3.5 Preformed or nonpreformed wire ropes. Preformed wire ropes are wire ropes in which the wires in the strands and the strands in the ropes were, prior to the last closing operation, permanently shaped to the spiral form (helix) they assume in the strands and wire ropes. Nonpreformed wire ropes are wire ropes which are fabricated from straight wires and strands. Preformed wire ropes are more flexible and less apt to unlay when cut from the reel.

6.3.6 Reels, manufacturing. Manufacturing reels are the reels on which the wire ropes are wound as they are fabricated on the rope closing machines.

6.3.7 Reels, shipping. Shipping reels are wooden reels on which the wire ropes, cut to the ordered lengths, are wound from the manufacturing reel for shipment.

6.3.8 Seize (or serve). To bind securely the ends of the wire ropes with a soft annealed iron wire (seizing wire).

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